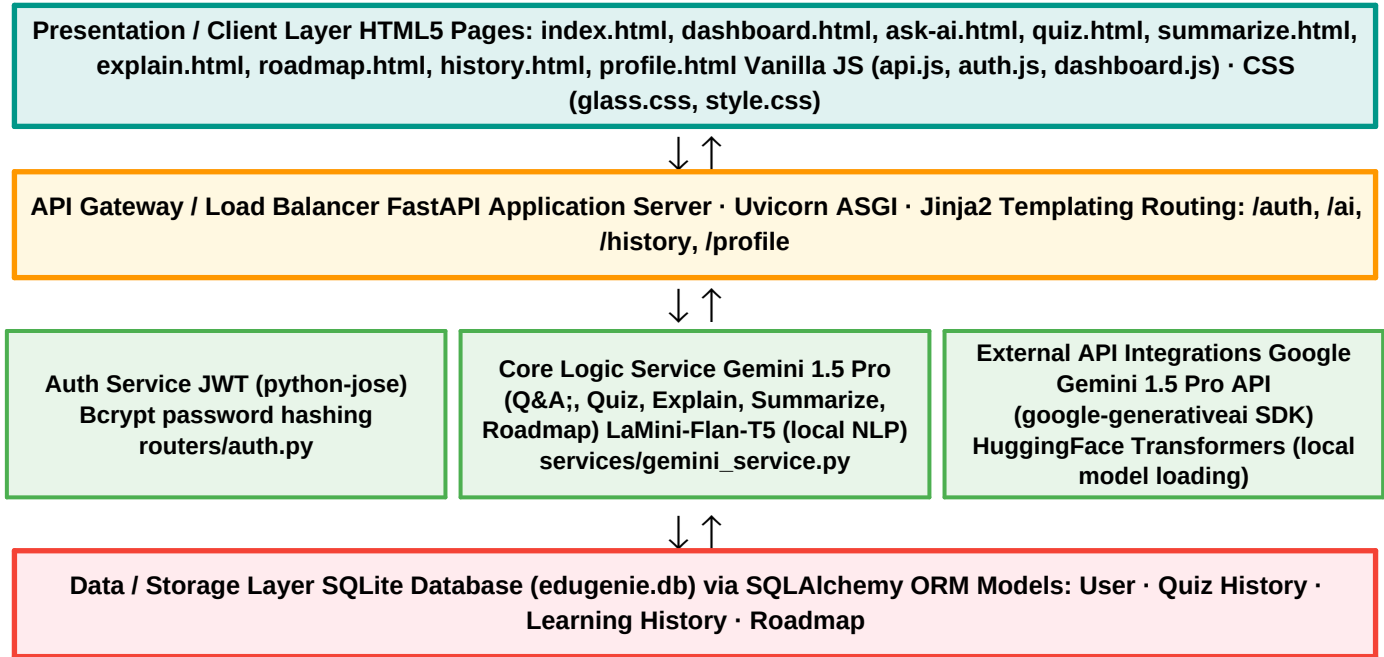


Solution Architecture

Date	26 June 2026
Team ID	PNT2026TMID-EG01
Project Name	EduGenie - AI-Powered Educational Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level visual representation of EduGenie's architecture across the Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Serves the user interface; all student-facing HTML pages for Q&A;, quiz, summary, roadmap, dashboard, auth. Communicates with backend via Fetch API (api.js).	HTML5, CSS3, Vanilla JavaScript, Jinja2
API Gateway (FastAPI)	Central request router receiving all HTTP requests from the frontend. Routes traffic to the correct service handler (auth, AI, history, profile). Handles CORS, middleware, and error responses.	FastAPI, Uvicorn ASGI, Python
Auth Service	Manages user registration, login, and JWT token issuance/validation. Protects all user-specific endpoints via dependency injection.	JWT (python-jose), Bcrypt, FastAPI
Core Logic Service (AI)	Processes all AI-powered educational requests: question answering, concept explanation, quiz generation, summarization, and learning roadmap. Interfaces with Gemini API and local LaMini-Flan-T5 model.	Google Gemini 1.5 Pro API, LaMini-Flan-T5, HuggingFace Transformers, Torch

External API Integration	Handles outbound API calls to Google Gemini 1.5 Pro for cloud-based AI generation and loads LaMini-Flan-T5 locally for lightweight NLP tasks.	google-generativeai SDK, HuggingFace Transformers
Data / Storage Layer	Persists all application data including user accounts, quiz history, learning history, and roadmap sessions. Accessed by all service layers via SQLAlchemy ORM.	SQLite (edugenie.db), SQLAlchemy ORM, Python